

previous sections. The Fama–French model (7.2) prices value stocks like Huffet and growth stocks like Inron relative to the market.

One important assumption in the CAPM and Fama–French models is that the betas are constant. Empirical evidence shows that exposures of some assets to systematic factors vary over time and, in particular, increase during bad times.²⁰ The variation of betas themselves can be a source of risk. That betas tend to increase during bad times undoubtedly caused the negative returns of risky assets to be larger during the financial crisis than they otherwise would have been had their betas remained constant.

3.2. SIZE FACTOR

The size effect was discovered by Banz (1981), with similar results in Reinganum (1981), and refers to the fact that small stocks tended to do better than large stocks, after adjusting for their betas. The past tense is appropriate here, because since the mid-1980s there has not been any significant size effect.

Figure 7.6 plots the value of \$1 invested in the *SMB* strategy after taking out the market effect beginning in January 1965 to December 2011 in the solid line.²¹

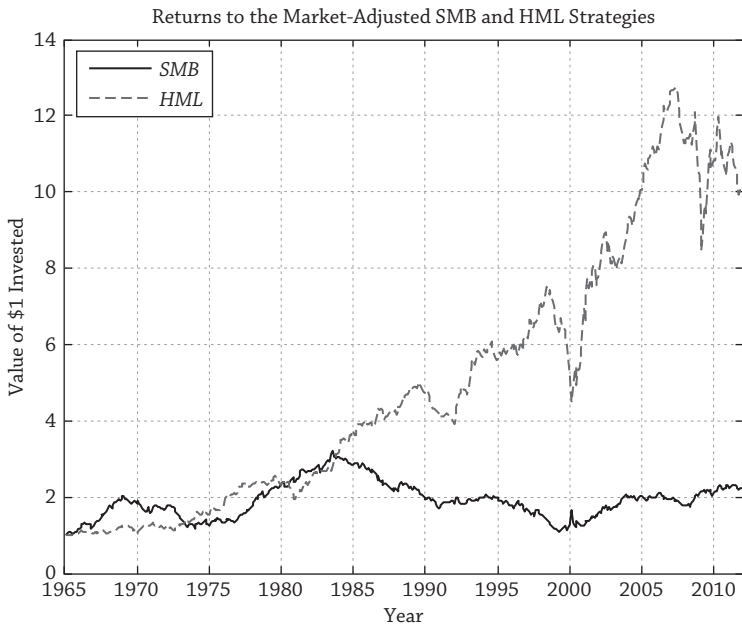


Figure 7.6

²⁰ See Ang and Chen (2002).

²¹ *SMB*, *HML*, and *WML* data for Figures 7.6 and 7.7 are from http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html.